

EduCollab Benchmark

Code Generation Prompts



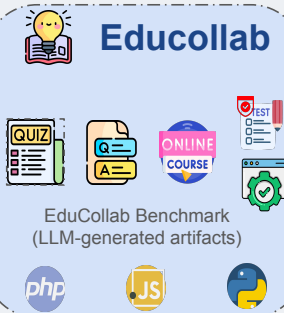
Project-level Prompt



Requirement-level Prompt



LLM Code Generation



Educollab

EduCollab Benchmark
(LLM-generated artifacts)

Vul. Detection and Vul. Repair Prompts

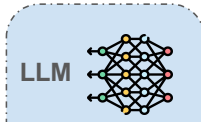
Project-level Prompt

Requirement-level Prompt

File-level Prompt



Vulnerability Detection



Vulnerability Repair

Iteration cap:
 $K(=1,2,3,4)$

Verification



(Functional + Exploit Tests)

Iterative DRV

Observations

Observation 1 : Marginal Gains Vary Across Repair Iterations

Additional repair iterations produce further secure-and-correct outcomes, but the marginal gains vary across iterations and analysis settings.

Observation 2 : Structural Completeness Does Not Establish Detection Accuracy

Structurally complete and detailed detection reports do not establish that the reported vulnerabilities are correct.

Observation 3 : Verification Outcomes Do Not Isolate Pipeline Failure Sources

End-to-end verification identifies unsuccessful outcomes but does not determine whether they result from detection errors, repair failures, or both.

Correct and Secure
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